SPORTS & ARTHROSCOPY · NEET-SS / FET / INI-SS

Sports medicine and anti-doping

1. Muscle injury and healing

- **Muscle strain:** commonest at the **myotendinous junction**, in **two-joint muscles** with **eccentric contraction** (**hamstrings (biceps femoris long head commonest)**, rectus femoris, medial gastrocnemius ("tennis leg"), adductor longus).
- **Healing phases: destruction/inflammation (days 1-3)**, **repair (regeneration of myofibres from satellite cells + scar formation, days 3-21)**, **remodelling** (weeks-months). **Satellite cells** = muscle stem cells.
- **Grading: grade I** (stretch, under 5% fibres), **II** (partial tear), **III** (complete rupture); **British Athletics Muscle Injury Classification (BAMIC)**: **0-4** with **a (myofascial), b (musculotendinous junction), c (intratendinous: longer return to play, higher recurrence)**.
- **Hamstring injuries:**
  - **Proximal hamstring avulsion** (**forced hip flexion with knee extension: water skiing**): **surgical repair** for **complete avulsion of 3 tendons**, or **2 tendons with over 2 cm retraction** in active patients; **early repair (within about 4-6 weeks)** gives better outcomes; **sciatic nerve** proximity.
  - **Prevention: Nordic hamstring eccentric programme** (reduces injuries by about 50%).
- **Treatment of muscle strains: PEACE & LOVE** (protection, elevation, avoid anti-inflammatories early, compression, education; load, optimism, vascularisation, exercise) has largely replaced **RICE/PRICE**; early controlled mobilisation; avoid immobilisation over a few days.
- **Delayed-onset muscle soreness (DOMS): 24-72 hours after unaccustomed eccentric exercise**; microtrauma; **raised CK**; self-limiting.

- **Myositis ossificans** (after contusion, **quadriceps**): **zonal pattern (mature peripheral ossification, immature centre)** (vs osteosarcoma: **reverse zoning**); **avoid massage/aggressive stretching early**; excise only after maturation (**about 6-12 months, bone scan cold**) if symptomatic.
- **Exertional rhabdomyolysis: CK over 5 times normal (often over 10,000)**, **myoglobinuria (dark urine)**, **acute kidney injury**; **sickle cell trait**, heat, dehydration; **aggressive IV fluids**.

2. Tendon injury and tendinopathy

- **Tendon structure: type I collagen (about 95%)**, **tenocytes**, proteoglycans (**decorin**); hierarchy: fibril, fibre, fascicle, tendon; **endotenon/epitenon/paratenon (vascular, in tendons without sheaths)**.
- **Tendon healing:** inflammatory (days), proliferative (**type III collagen**, weeks), remodelling (**type III replaced by type I**, months); **never reaches original strength (about 70-80%)**.
- **Tendinopathy: failed healing response** (**not inflammation: "tendinosis"**): **disorganised collagen, increased ground substance, neovascularisation with nerve ingrowth (pain)**. **Continuum model (Cook and Purdam): reactive tendinopathy, tendon disrepair, degenerative tendinopathy**.
- **Treatment: load management, eccentric (Alfredson) or heavy slow resistance programmes (best evidence)**; **isometrics for pain relief**; **ESWT**; **PRP** (inconsistent); **steroid injections around load-bearing tendons (Achilles, patellar): risk of rupture: avoid intratendinous**.
- **Fluoroquinolones** (ciprofloxacin): **tendinopathy/rupture (Achilles)**, especially **older patients, with corticosteroids, renal impairment**. **Statins** (possible).

3. Bone stress injuries

- **Stress fractures** (**fatigue fractures in normal bone** from repetitive loading) vs **insufficiency fractures** (normal load on abnormal bone).
- **Risk factors: sudden increase in training**, **female athlete triad/RED-S**, **low vitamin D**, **low BMI**, **military recruits**, **cavus feet**, leg-length discrepancy, hard surfaces, poor footwear.
- **Imaging:** radiographs often normal initially (**periosteal reaction after 2-3 weeks**); **MRI (most sensitive and specific; Fredericson grading 1-4)**; bone scan (sensitive, non-specific).
- **High-risk stress fractures (tension side, poor blood supply: prone to non-union/displacement: often surgery or strict non-weight-bearing):**
  - **Femoral neck (tension: superolateral)**,
  - **Anterior cortex of the mid-tibia ("dreaded black line")** (**jumpers**; IM nailing if not healing),
  - **Medial malleolus**,
  - **Talus (lateral process)**,
  - **Navicular (central third: avascular)**: **CT**; **non-weight-bearing cast 6-8 weeks** or **screw fixation** in athletes,
  - **Proximal fifth metatarsal (Jones/zone 2-3: metaphyseal-diaphyseal junction)**: **intramedullary screw** in athletes,
  - **Sesamoids of the hallux**,
  - **Patella**,
  - **Base of the second metatarsal (ballet dancers)**.
- **Low-risk (compression side): posteromedial tibia (commonest stress fracture overall)**, **metatarsal shafts (2nd and 3rd: "march fracture")**, femoral shaft, **fibula**, calcaneus, **femoral neck compression side (inferomedial)**: **relative rest, modified activity**.
- **Medial tibial stress syndrome ("shin splints")**: **diffuse posteromedial tibial pain (distal two-thirds)**, **periostitis/traction**

of the soleus/FDL/tibialis posterior; **MRI: periosteal oedema without fracture line**; rest, gradual return, **correct training errors**.

- **Chronic exertional compartment syndrome (CECS): anterior compartment commonest (then deep posterior)**; **pain with exercise, relieved by rest, ± numbness/foot drop**; **compartment pressure testing**: **(Pedowitz criteria) pre-exercise ≥15 mmHg, 1 minute post-exercise ≥30 mmHg, or 5 minutes post-exercise ≥20 mmHg**; **fasciotomy** (anterior and lateral; **superficial peroneal nerve** at risk where it pierces the fascia about **10-12 cm above the lateral malleolus**).
- **Popliteal artery entrapment syndrome** (young athletes: claudication; **medial head of gastrocnemius anomaly**), **external iliac artery endofibrosis** (cyclists): differentials of exertional leg pain.

4. Female athlete triad / RED-S

- **Female athlete triad (ACSM): low energy availability (with or without disordered eating), menstrual dysfunction (functional hypothalamic amenorrhoea), low bone mineral density** (**each on a spectrum**).
- **Relative Energy Deficiency in Sport (RED-S, IOC 2014; updated 2023)**: broader concept affecting **both sexes**, multiple systems (**metabolic, haematological, cardiovascular, psychological, immunological, growth and development, performance**).
- **Low energy availability: under about 30 kcal/kg fat-free mass/day.**
- **Bone: low BMD (Z-score under -1.0 in athletes is concerning; under -2.0 = low BMD for age)** (use **Z-scores**, not T-scores, in premenopausal women and young athletes), **stress fractures (trabecular sites: femoral neck, sacrum, pelvis)**.
- **Treatment: multidisciplinary** (sports physician, dietitian, psychologist); **increase energy intake/reduce training**: restoration

of menses; **oral contraceptive pills do not restore BMD** (mask amenorrhoea); **transdermal oestrogen** may be used if needed; calcium and vitamin D.

5. Concussion (sport-related)

- **Traumatic brain injury induced by biomechanical forces**: **functional rather than structural** (normal standard neuroimaging); symptoms may evolve over hours.
- **Signs:** loss of consciousness (**under 10%**), **amnesia**, confusion, headache, dizziness, balance problems, visual disturbance, **"just not right"**.
- **Assessment: SCAT6 (Sport Concussion Assessment Tool; Amsterdam 2022 consensus)** for athletes 13 years and older; **Child SCAT6** for 8-12 years; **Concussion Recognition Tool (CRT6)** for non-medical people.
- **"If in doubt, sit them out": no return to play on the same day.**
- **Red flags (immediate removal and emergency referral):** neck pain/tenderness, **double vision**, weakness/tingling in limbs, **severe or increasing headache**, **seizure**, **loss of consciousness**, **deteriorating conscious state**, **vomiting**, increasingly restless/agitated/combative.
- **Return to sport (graduated, stepwise): relative rest for 24-48 hours** (light activity encouraged), then **symptom-limited activity, aerobic exercise, sport-specific exercise, non-contact training drills, full-contact practice (after medical clearance), return to sport**; **at least 24 hours per step**; **children/adolescents: more conservative; return to learn before return to sport**.
- **Second impact syndrome: diffuse cerebral swelling** after a second concussion before recovery from the first: **adolescents**; often fatal.
- **Chronic traumatic encephalopathy (CTE):** tauopathy associated with repetitive head impacts (contact sports).

6. Pre-participation evaluation and cardiac issues

- **Sudden cardiac death in young athletes: hypertrophic cardiomyopathy (commonest in US athletes)**, **arrhythmogenic right ventricular cardiomyopathy (Italy)**, **coronary artery anomalies**, **myocarditis**, **Marfan syndrome (aortic dissection)**, **commotio cordis** (blunt chest impact during repolarisation: **ventricular fibrillation**; **immediate CPR + AED**), **long QT**, **Brugada**, **WPW**.
- **Screening:** history and examination (AHA 14-point) ± **ECG (European/IOC practice)**.
- **Athlete's heart (physiological)**: sinus bradycardia, **first-degree AV block**, early repolarisation, isolated voltage criteria for LVH: **normal ECG variants in athletes (international criteria)**.
- **Exercise-induced bronchoconstriction**, **exertional heat stroke** (**core temperature over 40 C + CNS dysfunction**: **cold-water immersion ("cool first, transport second")**), **hyponatraemia (exercise-associated: over-drinking hypotonic fluids in endurance events)**.
- **Marfan:** avoid contact and isometric sports (aortic root dilatation).
- **Atlantoaxial instability in Down syndrome** (screening for contact sports, Special Olympics).
- **Return to play after infectious mononucleosis: splenic rupture risk (days 4-21)**: **avoid contact sports for at least 3-4 weeks** (until splenomegaly resolves).

7. Anti-doping

Organisation

- **World Anti-Doping Agency (WADA)**: founded **1999** (after the 1998 Tour de France Festina scandal); publishes the **World Anti-Doping Code** and the **Prohibited List** (updated **annually, effective 1 January**).

- **National anti-doping organisations: India: NADA (National Anti-Doping Agency, established 2005)**; **National Dope Testing Laboratory (NDTL), New Delhi**; **National Anti-Doping Act, 2022**.
- **Court of Arbitration for Sport (CAS)**: appeals.

Criteria for inclusion on the Prohibited List (any two of three)

1. **Potential to enhance sport performance.**
2. **Actual or potential health risk to the athlete.**
3. **Violates the spirit of sport.**
4. (Or masks the use of other prohibited substances.)

Prohibited List categories

- **Prohibited at all times (in and out of competition):**
  - **S0 Non-approved substances**,
  - **S1 Anabolic agents** (**anabolic androgenic steroids**: testosterone, **nandrolone**, **stanozolol**; **other anabolic agents: clenbuterol, SARMs, tibolone, zeranol**),
  - **S2 Peptide hormones, growth factors and related substances** (**EPO and erythropoiesis-stimulating agents**, **HIF activators** (e.g. **roxadustat**, **cobalt**), **hCG and LH (in males)**, **growth hormone and its releasing factors**, **IGF-1**, corticotrophins),
  - **S3 Beta-2 agonists** (all prohibited, **except inhaled salbutamol (max 1,600 µg/24 h, not exceeding 600 µg over 8 hours)**, **inhaled formoterol (max 54 µg/24 h)**, **inhaled salmeterol (max 200 µg/24 h)**, **inhaled vilanterol (max 25 µg/24 h)** within therapeutic doses),
  - **S4 Hormone and metabolic modulators** (**aromatase inhibitors** (anastrozole, letrozole), **SERMs** (tamoxifen, clomiphene), **myostatin inhibitors**, **insulin and insulin mimetics**, **meldonium**, **trimetazidine**, **AICAR, GW1516 (PPAR-delta agonists)**),

  - **S5 Diuretics and masking agents** (**furosemide, thiazides, spironolactone, acetazolamide**, **desmopressin**, **probenecid**, **plasma expanders: albumin, dextran, HES**, **mannitol IV**) (**drospirenone, pamabrom, topical dorzolamide/brinzolamide are permitted**),
  - **M1 Manipulation of blood and blood components** (**blood doping: autologous, allogenic, homologous transfusion**; artificial oxygen carriers),
  - **M2 Chemical and physical manipulation** (**tampering, urine substitution**, **IV infusions/injections over 100 mL per 12-hour period** except during hospital treatment, surgery or clinical investigations),
  - **M3 Gene and cell doping**.
- **Prohibited in competition only:**
  - **S6 Stimulants** (amphetamines, cocaine, **ephedrine (over 10 µg/mL urine)**, **pseudoephedrine (over 150 µg/mL)**, methylhexaneamine, modafinil, **methylphenidate**, strychnine) (**caffeine, nicotine, phenylephrine, synephrine are on the monitoring programme, not prohibited**),
  - **S7 Narcotics** (morphine, oxycodone, fentanyl, methadone, pethidine, **tramadol (prohibited in competition since 2024)**) (**codeine permitted**),
  - **S8 Cannabinoids** (**except cannabidiol (CBD)**; THC urinary threshold **150 ng/mL**),
  - **S9 Glucocorticoids** (**all injectable routes, oral, rectal** prohibited in competition; **inhaled, topical, intranasal are permitted**; **washout periods** for injections, e.g. about 3 days intramuscular/intra-articular for most).
- **Prohibited in particular sports: P1 Beta-blockers** (**archery, shooting (also out of competition)**, golf, billiards, darts, motorsport, skiing (some disciplines), underwater sports).

- **Local anaesthetics (lidocaine), NSAIDs, paracetamol, insulin for diabetes (with TUE), vitamins** are not prohibited (insulin requires TUE).

Anti-doping rule violations (ADRVs: 11 under the Code)

- **Presence of a prohibited substance** in a sample (**strict liability: the athlete is responsible for any substance found, regardless of intent**), use/attempted use, **evading/refusing sample collection**, **whereabouts failures (3 within 12 months)**, tampering, possession, trafficking, administration, complicity, prohibited association, **intimidating/retaliating against whistle-blowers**.
- **Sanctions: standard 4 years** for intentional violations involving non-specified substances; **2 years** if not intentional; **reduced for "no significant fault"** or **contaminated products**; **lifetime ban** for repeat serious violations.
- **Athlete Biological Passport (ABP): longitudinal monitoring of biomarkers** (haematological module: haemoglobin, reticulocytes, **OFF-score**; steroidal module: **testosterone/epitestosterone (T/E) ratio (over 4 suspicious)**; endocrine module (GH)); detects doping indirectly.
- **Testing methods:** urine and blood; **isotope ratio mass spectrometry (IRMS: exogenous testosterone)**, **GC-MS/LC-MS**, **dried blood spots**.

Therapeutic Use Exemption (TUE)

- **Criteria (all must be met):**
  - **Significant health impairment** without the substance,
  - **No more than return to normal health** (no additional performance enhancement),
  - **No reasonable permitted therapeutic alternative**,
  - Not a consequence of prior non-therapeutic use of a prohibited substance.

- **Apply in advance** (retroactive TUE for emergencies). Examples: **insulin (diabetes)**, **glucocorticoids (systemic)**, **methylphenidate (ADHD)**, **testosterone (genuine hypogonadism)**, **beta-blockers in shooting (cardiac indication)**.

Doctor's responsibilities

- **Check every medication against the current Prohibited List** (e.g. **Global DRO**, NADA app), **record medications**, **advise on supplements (contamination risk: anabolic agents/stimulants in supplements)**, **apply for TUE**, **IV infusion limits**; **doctors can be sanctioned** (administration, complicity).

8. Other sports medicine topics

- **Platelet-rich plasma (PRP)**: autologous platelet concentrate (**growth factors: PDGF, TGF-beta, VEGF**); **not prohibited by WADA** (growth factors within PRP: **permitted since 2011**); evidence best for **knee OA and lateral epicondylitis** (mixed).
- **Exercise prescription: WHO 2020: adults 150-300 minutes moderate or 75-150 minutes vigorous aerobic activity per week + muscle-strengthening 2 or more days/week**.
- **Athletes and NSAIDs:** may impair **bone and tendon-bone healing** (fracture non-union concern), **renal injury with dehydration**.
- **Sports drinks and carbohydrate loading**; **creatine** (not prohibited; increases phosphocreatine stores: short-burst power).

> Exam pearl: **WADA (1999) Prohibited List annual (1 January); India: NADA (2005), NDTL, National Anti-Doping Act 2022. Inclusion = any 2 of 3: performance, health risk, spirit of sport. Strict liability. Standard intentional sanction 4 years. Whereabouts: 3 failures in 12 months.**

Exam pearl: **At all times: anabolic agents, EPO/HIF activators, GH, beta-2 agonists (except inhaled salbutamol under 1,600 µg/day, formoterol, salmeterol), aromatase inhibitors/SERMs, insulin, meldonium, diuretics/masking agents, blood doping, IV infusion over 100 mL/12 h, gene doping. In competition: stimulants, narcotics (tramadol since 2024; codeine allowed), cannabinoids (CBD allowed), systemic/injected glucocorticoids. Beta-blockers in archery/shooting. TUE: health impairment, no enhancement, no alternative.**

Exam pearl: **High-risk stress fractures: femoral neck tension side, anterior tibia (dreaded black line), navicular, Jones (5th MT proximal), medial malleolus, sesamoids. RED-S: use Z-scores; OCP does not restore BMD. Concussion: no same-day return, SCAT6, stepwise return at least 24 h per step. Commotio cordis: AED.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.